ANUSHANGA KAUSHAN MUNASINGHE

Software Developer & IT Undergraduate

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LinkedIn Profile | Portfolio Website | GitHub

SUMMARY

Motivated and detail-oriented IT undergraduate at SLIIT with practical experience as an IT & Full-stack Developer Intern at Sri Lanka Customs. Proven track record in full-stack software development, machine learning integration, and secure enterprise desktop applications. Experienced in independently designing, building, and deploying production-grade systems for government institutions and multi-disciplinary research projects. Seeking to leverage technical skills to build scalable solutions and contribute value as a software engineering professional.

EXPERIENCE

Software Developer (Contract / Project-based) — Sri Lanka Customs (Post-Internship / Present)

- Architected and independently delivered the enterprise Seizure Report (SR) Management System as a department-wide solution for Customs operations across Sri Lanka, providing ongoing system maintenance and post-deployment management (detailed in Projects).

IT & Full-stack Developer Intern — Sri Lanka Customs (Internship Period)

- Designed and developed enterprise desktop and departmental solutions including the Library Management System (SLC-LMS) and Learning Management System (SLCHRD LMS).
- Implemented secure database systems with row-level security, role-based access control (RBAC), and dynamic reporting engines for government workflows.

TECHNICAL SKILLS

- **Programming Languages & ML:** (FastAPI, Scikit-Learn, Random Forest, XGBoost), Dart (Flutter), JavaScript, TypeScript, Java, PHP, C, C++
- **Frameworks & Libraries:** FastAPI, React Native, Next.js, Electron.js, Node.js, Spring Boot, MERN Stack, Java Swing, CustomTkinter, Tailwind CSS, Bootstrap, Material UI
- **Databases & Architecture:** PostgreSQL 18 (RLS, Triggers, Migrations), MongoDB, SQLite3 (WAL), Direct DB Driver Architecture
- **Tools & DevOps:** Git/GitHub, Postman, Figma, Android Studio, VS Code, Inno Setup, PDF/Excel Generation Libraries

KEY PROJECTS & RESEARCH

- **SR Management System (Sri Lanka Customs):** Engineered a comprehensive, enterprise-grade multi-portal Windows desktop system in Flutter and PostgreSQL 18 managing the complete seizure-to-disposal lifecycle as a department-wide solution for all yards across Sri Lanka (active validation at GL1, GL2, NNR, RCT). Implemented 153 Row-Level Security (RLS) policies, 7 role-based portals, parameterised queries, background isolate processing, and independent security hardening.
- **Smart Fisher Lanka - Fish Price Prediction & Market Intelligence Engine (Research Module):** Developed a hybrid machine learning ensemble model (Random Forest & XGBoost) achieving 88.7% **R^2** accuracy to forecast fish prices for 32 species across Sri Lanka. Engineered a FastAPI backend integrating 33 features, a 31-day trend forecasting engine, Explainable AI (XAI) reason cards, and a Smart Fish Recommendation Engine.
- **Ambalantota Pradeshiya Sabha - Library Management System:** Engineered a production-grade desktop application using Electron.js, Node.js, and PostgreSQL (Neon Tech). Features role-based authentication with bcryptjs and Gmail SMTP OTP, copy-level accession tracking, circulation management, and automated PDF (pdfkit) and Excel (exceljs) reporting with auto-updates via GitHub Releases
- **Sri Lanka Customs - Library Management System (SLC-LMS):** Developed a secure desktop information system using Python, CustomTkinter, and SQLite3 in WAL mode with role-based access control (RBAC), PBKDF2-HMAC-SHA256 hashing, and automated fine computation.
- **Sri Lanka Customs HRD - Learning Management System (SLCHRD LMS):** Built an enterprise 3-tier desktop application using Java Swing, FlatLaf, OpenJDK 24, and MySQL 8.x with secure authentication and Apache PDFBox reporting
- **Modern Social Media Application:** Developed using Spring Boot, MongoDB, Spring Data JPA (Hibernate), React JS, Ant Design, and Tailwind CSS. Tools used: Postman and GitHub.
- **Smart Home Stock and Sharing Management System:** Developed using MERN Stack (MongoDB, Express.js, React.js, Node.js) and Tailwind CSS, utilizing Postman and GitHub.
- **Online Booking Tour Web Application:** Developed a 2nd year, 1st semester OOP project using HTML, CSS, JavaScript, Java, and SQL (JSP / Java Servlets).
- **Rukshan Furniture E-Commerce Web Application:** Developed a full-stack web application using the MERN stack (MongoDB, Express.js, React.js, Node.js) featuring file/image upload functionalities, an integrated online chat box, and custom CSS styling.

EDUCATION

- **B.Sc. (Hons) in Information Technology —** Sri Lanka Institute of Information Technology (SLIIT) (Undergraduate)
- **G.C.E. Advanced Level (2020) —** S.F.T. (C), I.T. (C), E.T. (C), General English (S)
- **G.C.E. Ordinary Level (2017) —** Passed with 3A, 4B, 1C,S

CERTIFICATIONS

- **Java Programming Certificate -** Dreamteam IT Solutions
- **Python for Beginners -** University of Moratuwa
- **Web Design for Beginners -** University of Moratuwa
- **Java & HTML Sololearn Certificates**

REFERENCES

Prof. Pradeep Abeygunawardhana
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